Insurance

Homework #4

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1 Introduction

We model two outcomes in a auto insurance outcome variables: (1) whether the person was in a car crash (binary) and (2) the amount of money it will cost if the person does crash their car (continuous). Our objective is to build multiple linear regression and binary logistic regression models on the training data to predict the probability that a person will crash their car and also predict the cost of the crash if it occurs.

2 Data Exploration and Cleaning

```r
# Load the training and evaluation datasets
insurance_train <- read_csv("./data/insurance_training_data.csv")
insurance_eval <- read_csv("./data/insurance-evaluation-data.csv")

# Count rows and columns
nrows <- insurance_train |> nrow()
ncols <- insurance_train |> length()

# Get missingness
perc_complete <- round(insurance_train |> n_complete() / nrows / ncols * 100, 2)
```

The insurance training data set contains 8161 rows and 26 features, and is 98.87 percent complete. We will need to address the missingness in the data preparation section.

There are several variables in the dataset coded as characters. There are several currency variables (`INCOME`, `HOME_VAL`, `BLUEBOOK`, `OLDCLAIM`) that should be changed to numeric variables. The rest of the character variables should be encoded as dummy variables for modeling in the following chunk. We also rename some variables for clarity. We also remove the `index` variable as it is not a predictor and will not be used in modeling.

```
# Convert currency variables to numeric by removing "$" and converting to numeric
currency_vars <- c("INCOME", "HOME_VAL", "BLUEBOOK", "OLDCLAIM")

# Character columns to one-hot encode (exclude currency columns)
char_vars <- insurance_train |>
    select(where(is.character)) |>
    names() |>
    setdiff(currency_vars)

insurance_train_dummies <- insurance_train |>
    mutate(
        # Convert currency variables to numeric
        across(all_of(currency_vars), parse_number)
    ) |>
    fastDummies::dummy_cols(
        select_columns = char_vars,
        remove_selected_columns = TRUE,
        remove_first_dummy = TRUE
    ) |>
    clean_names() |>
    rename(
        female = sex_z_f,
        education_phd = education_ph_d,
        education_high_school = education_z_high_school,
        car_type_suv = car_type_z_suv,
        job_blue_collar = job_z_blue_collar,
        urbanicity_highly_rural = urbanicity_z_highly_rural_rural
    ) |>
    select(-index)

insurance_eval_dummies <- insurance_eval |>
    mutate(
        across(all_of(currency_vars), parse_number)
    ) |>
    fastDummies::dummy_cols(
        select_columns = char_vars,
        remove_selected_columns = TRUE,
```

```
        remove_first_dummy = TRUE
    ) |>
    clean_names()|>
    rename(
        female = sex_z_f,
        education_phd = education_ph_d,
        education_high_school = education_z_high_school,
        car_type_suv = car_type_z_suv,
        job_blue_collar = job_z_blue_collar,
        urbanicity_highly_rural = urbanicity_z_highly_rural_rural
    ) |>
    select(-index)
```

In previewing summary statistics for all variables, some of the minimum values appear to be in error. For example, `car_age` has a negative value, and there are a high number of 0 values in `car_age`, `home_val`, which may be data entry errors–we will replace them missing values.

```
insurance_train_dummies |> summary()

insurance_train_dummies <- insurance_train_dummies |>
    mutate(
        car_age = ifelse(car_age <= 0, NA, car_age),
        home_val = ifelse(home_val == 0, NA, home_val)
    )
```

Table 1: Summary statistics for all variables in the training dataset

Variable	Completeness rate	Mean	Min	P25	P50	P75	Max
target_flag	1.00	0.260	0	0	0	1	1
target_amt	1.00	1500.000	0	0	0	1000	110000
kidsdriv	1.00	0.170	0	0	0	0	4
age	1.00	45.000	16	39	45	51	81
homekids	1.00	0.720	0	0	0	1	5
yoj	0.94	10.000	0	9	11	13	23
income	0.95	62000.000	0	28000	54000	86000	370000
home_val	0.66	220000.000	50000	150000	210000	270000	890000
travtime	1.00	33.000	5	22	33	44	140
bluebook	1.00	16000.000	1500	9300	14000	21000	70000
tif	1.00	5.400	1	1	4	7	25
oldclaim	1.00	4000.000	0	0	0	4600	57000

Table 1: Summary statistics for all variables in the training dataset

Variable	Completeness rate	Mean	Min	P25	P50	P75	Max
clm_freq	1.00	0.800	0	0	0	2	5
mvr_pts	1.00	1.700	0	0	1	3	13
car_age	0.94	8.300	1	1	8	12	28
parent1_yes	1.00	0.130	0	0	0	0	1
mstatus_z_no	1.00	0.400	0	0	0	1	1
female	1.00	0.540	0	0	1	1	1
education_bachelors	1.00	0.270	0	0	0	1	1
education_masters	1.00	0.200	0	0	0	0	1
education_phd	1.00	0.089	0	0	0	0	1
education_high_school	1.00	0.290	0	0	0	1	1
job_doctor	0.94	0.032	0	0	0	0	1
job_home_maker	0.94	0.084	0	0	0	0	1
job_lawyer	0.94	0.110	0	0	0	0	1
job_manager	0.94	0.130	0	0	0	0	1
job_professional	0.94	0.150	0	0	0	0	1
job_student	0.94	0.093	0	0	0	0	1
job_blue_collar	0.94	0.240	0	0	0	0	1
job_na	1.00	0.064	0	0	0	0	1
car_use_private	1.00	0.630	0	0	1	1	1
car_type_panel_truck	1.00	0.083	0	0	0	0	1
car_type_pickup	1.00	0.170	0	0	0	0	1
car_type_sports_car	1.00	0.110	0	0	0	0	1
car_type_van	1.00	0.092	0	0	0	0	1
car_type_suv	1.00	0.280	0	0	0	1	1
red_car_yes	1.00	0.290	0	0	0	1	1
revoked_yes	1.00	0.120	0	0	0	0	1
urbanicity_highly_rural	1.00	0.200	0	0	0	0	1

Table 1 presents descriptive characteristics of the training data. The target variables are `target_flag`, which represents whether the person was in a car crash (1) or not (0), and `target_amt`, which represents the cost of the crash if it occurs. The other variables represent various information on auto insurance for different individuals.

```
# Get correlation matrix
insurance_train_cor <- insurance_train_dummies |>
    select(where(is.numeric)) |>
```

```
    cor(use = "pairwise.complete.obs")

insurance_train_cor_plot <- insurance_train_cor
insurance_train_cor_plot[abs(insurance_train_cor_plot) < 0.2] <- NA

corrplot(insurance_train_cor_plot, method = "circle", type = "upper", sig.level = 0.05, insig

most_corr_w_target_flag <- insurance_train_cor |>
    as.data.frame() |>
    rownames_to_column(var = "Variable") |>
    filter(Variable != "target_flag", Variable != "target_amt") |>
    transmute(
        Variable,
        Correlation = target_flag,
        Abs_Correlation = abs(target_flag)
    ) |>
    arrange(desc(Abs_Correlation)) |>
    filter(Abs_Correlation > 0.2) |> pull(Variable)

most_corr_w_target_amt <- insurance_train_cor |>
    as.data.frame() |>
    rownames_to_column(var = "Variable") |>
    filter(Variable != "target_flag", Variable != "target_amt") |>
    transmute(
        Variable,
        Correlation = target_amt,
        Abs_Correlation = abs(target_amt)
    ) |>
    arrange(desc(Abs_Correlation)) |>
    filter(Abs_Correlation > 0.2) |>
    pull(Variable)
```

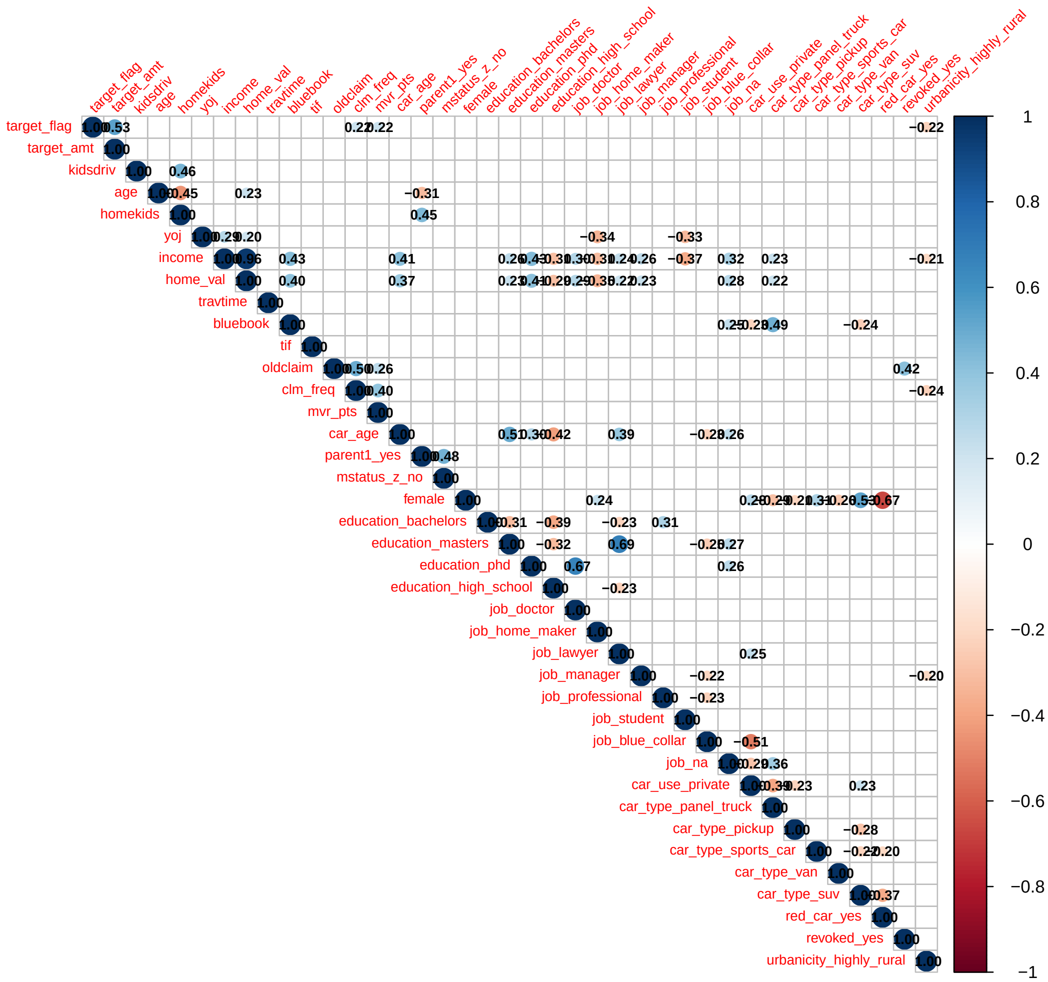

Figure 1: Correlation matrix of all variables in the training dataset. Note: Correlations with absolute value less than 0.2 are suppressed.

The correlation between all variables are shown Figure 1. The variables most correlated with `target_flag`, whether there was a car crash, (besides the amount the crash costed) are urbanicity_highly_rural, mvr_pts, clm_freq. This means that people who work or live in a more rural area are less likely to be in a crash, and people who have more motor vehicle record points or have more claims in the past 5 years are more likely to be in a crash. There were no predictor variables strongly correlated with `target_amt`, the cost if the crash occurs.

Lastly, for continuous variables, we also looked at the distribution of each variable to check for skewness and outliers.

```
# Keep only continuous
continuous_vars <- insurance_train_dummies |>
    select(where(is.numeric)) |>
    select(where(~ dplyr::n_distinct(stats::na.omit(.x)) > 10)) |>
    names()

insurance_train_dummies |>
    select(all_of(continuous_vars)) |>
    pivot_longer(cols = everything(), names_to = "variable", values_to = "value") |>
    ggplot(aes(x = value)) +
    geom_histogram(bins = 30, fill = "steelblue", color = "white") +
    facet_wrap(~ variable, scales = "free", ncol = 4) +
    theme_minimal() +
    theme(panel.grid = element_blank()) +
    labs(x = NULL, y = "Count")
```

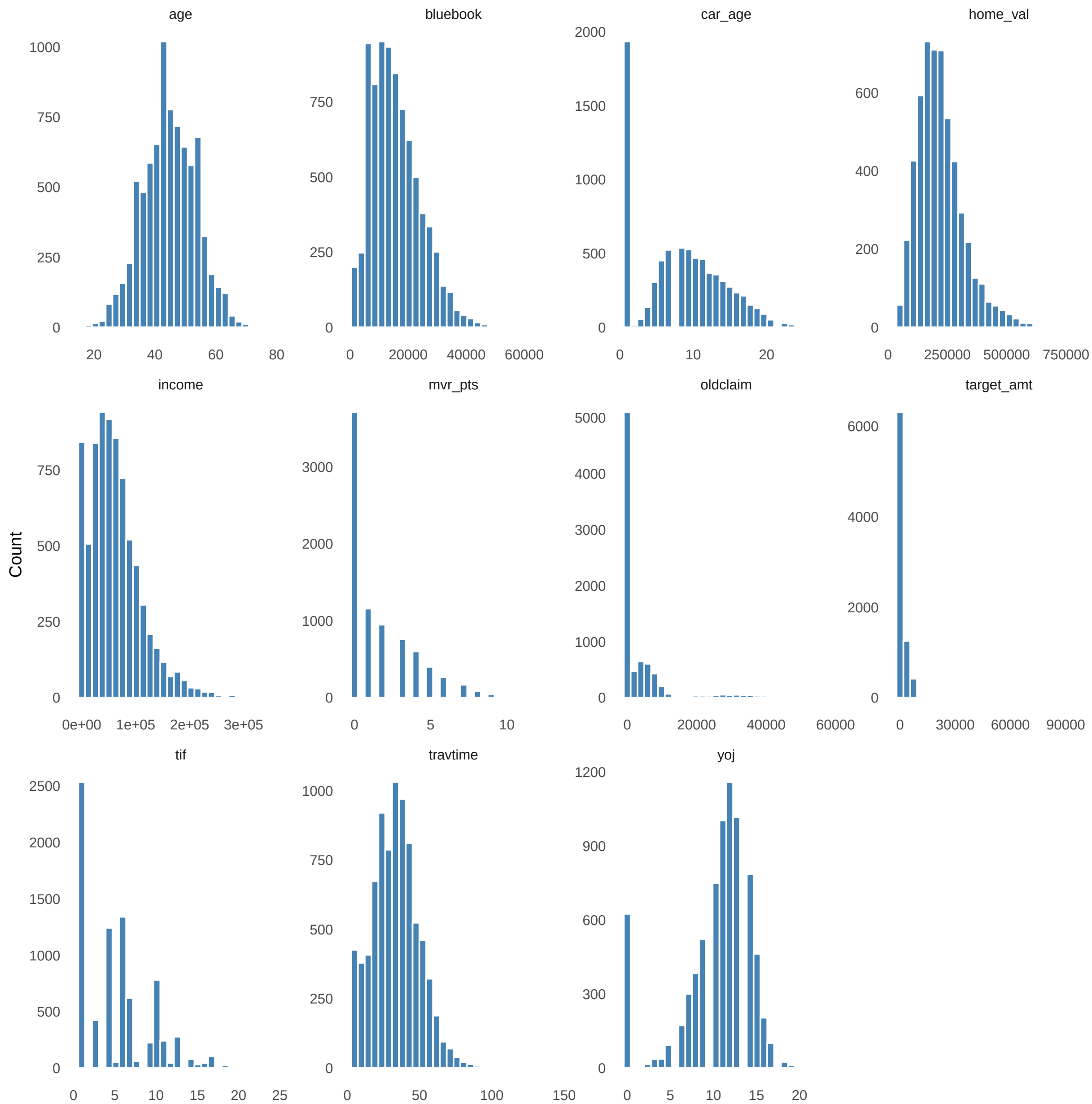

Figure 2: Distribution of continuous variables in the training dataset.

There are many 1-year old cars in the dataset, showing a high spike at 1 in the `car_age` variable. Dollar amounts such as `bluebook`, `home_val`, `income`, and `target_amt` are right-skewed, which is expected for financial variables. The same goes for travel time (`travtime`). Most people have no motor vehicle record points (`mvr_pts`) or claims in the past 5 years (`oldclaim`), which is expected for a general population. Years on the job (`yoj`) is almost normally distributed, but has a spike at 0, which may represent people who are unemployed or have been on the job for less than a year.

3 Data Preparation

The goal of this section is to prepare the insurance training and evaluation data for modeling. The assignment requires transformations of the original variables or the creation of new variables when useful. In this dataset, preparation included missing value treatment, missingness flags, mathematical transformations, and derived variables.

```
# Identify numeric predictor variables
numeric_vars <- insurance_train_dummies |>
  select(where(is.numeric)) |>
  select(-target_flag, -target_amt) |>
  names()

# Create missingness flags before imputation
insurance_train_prepared <- insurance_train_dummies |>
  mutate(
    age_missing = ifelse(is.na(age), 1, 0),
    car_age_missing = ifelse(is.na(car_age), 1, 0),
    home_val_missing = ifelse(is.na(home_val), 1, 0),
    income_missing = ifelse(is.na(income), 1, 0),
    yoj_missing = ifelse(is.na(yoj), 1, 0)
  )

insurance_eval_prepared <- insurance_eval_dummies |>
  mutate(
    age_missing = ifelse(is.na(age), 1, 0),
    car_age_missing = ifelse(is.na(car_age), 1, 0),
    home_val_missing = ifelse(is.na(home_val), 1, 0),
    income_missing = ifelse(is.na(income), 1, 0),
    yoj_missing = ifelse(is.na(yoj), 1, 0)
  )

# Median imputation using training medians
for (v in numeric_vars) {
  med_val <- median(insurance_train_prepared[[v]], na.rm = TRUE)

  insurance_train_prepared[[v]][is.na(insurance_train_prepared[[v]])] <- med_val
  insurance_eval_prepared[[v]][is.na(insurance_eval_prepared[[v]])] <- med_val
}
```

```
# Create transformed and engineered variables
insurance_train_prepared <- insurance_train_prepared |>
  mutate(
    log_income = log1p(income),
    log_home_val = log1p(home_val),
    log_bluebook = log1p(bluebook),
    log_oldclaim = log1p(oldclaim),
    log_travtime = log1p(travtime),

    claim_rate = oldclaim / (clm_freq + 1),
    car_value_age_ratio = bluebook / (car_age + 1),
    risk_points_claims = mvr_pts + clm_freq,

    age_bucket = case_when(
      age < 25 ~ "young_driver",
      age >= 25 & age < 60 ~ "adult_driver",
      age >= 60 ~ "older_driver",
      TRUE ~ "unknown"
    )
  )

insurance_eval_prepared <- insurance_eval_prepared |>
  mutate(
    log_income = log1p(income),
    log_home_val = log1p(home_val),
    log_bluebook = log1p(bluebook),
    log_oldclaim = log1p(oldclaim),
    log_travtime = log1p(travtime),

    claim_rate = oldclaim / (clm_freq + 1),
    car_value_age_ratio = bluebook / (car_age + 1),
    risk_points_claims = mvr_pts + clm_freq,

    age_bucket = case_when(
      age < 25 ~ "young_driver",
      age >= 25 & age < 60 ~ "adult_driver",
      age >= 60 ~ "older_driver",
      TRUE ~ "unknown"
    )
```

```
  )

# Convert age bucket into dummy variables
insurance_train_prepared <- fastDummies::dummy_cols(
  insurance_train_prepared,
  select_columns = "age_bucket",
  remove_selected_columns = TRUE,
  remove_first_dummy = TRUE
)

insurance_eval_prepared <- fastDummies::dummy_cols(
  insurance_eval_prepared,
  select_columns = "age_bucket",
  remove_selected_columns = TRUE,
  remove_first_dummy = TRUE
)

# Confirm no missing values remain
sum(is.na(insurance_train_prepared))
```

```
[1] 0
```

```
sum(is.na(insurance_eval_prepared))
```

```
[1] 4282
```

3.0.1 Linearity Assumption Summary

The results above indicate that the assumption of linearity is satisfied for all independent variables. However, several were flagged as being unlikely to have predictive value in a multivariate linear model.

3.1 Data Splits

First, we will split our dataset into a training and testing set. We will use the training set to build our models and the testing set to evaluate their performance. The training set will be used to fit the models, while the testing set will be used to assess how well the models generalize to new data.

```
set.seed(19940211)
train_indices <- sample(nrow(insurance_train_prepared), size = 0.8 * nrow(insurance_train_prepa
insurance_train_split <- insurance_train_prepared[train_indices, ]
insurance_test_split <- insurance_train_prepared[-train_indices, ]
```

::: {.callout notetidytable} RR Note: This whole assumption section needs to be updated :::

4 Check assumptions for binary logistic regression

1. Binary Outcome

```r
# Confirming that the target is binary
crime_clean |>
    count(target) |>
    mutate(prop = n / sum(n))
```

The response variable “target” is confirmed to be a binary, therefore it takes two values (0: crime rate below the median) and (1: crime rate above the median). In the 466 neighborhoods in training data set 237 (50.9%) fall in the class of low crime and 228 (49.1)% are in the high crime neighborhood. There is no class imbalance, due to the near equal split meaning there is unlikely to be bias towards one class.

2. Independence of Observations

Each record in the dataset represents a distinct neighborhood, and there is no repeated-measures or clustered structure in the data. Therefore, the independence of observations assumption is satisfied by design and requires no further adjustment.

3. No Multicollinearity (VIF Check)

```r
library(car)

# Fit a reference logistic model with original predictors
model_ref <- glm(
    target ~ zn + indus + chas + nox + rm + age + dis + rad + tax + ptratio
        + lstat + medv,
    data = crime_clean, family = binomial
)

# Compute VIF
vif_vals <- vif(model_ref) |>
    as.data.frame() |>
    rownames_to_column("Variable") |>
    rename(VIF = 2) |>
    arrange(desc(VIF))

vif_vals |> knitr::kable(caption = "Variance Inflation Factors for Reference Model")
```

The VIF values is calculated for all predictors to check for multicollinearity. A VIF above 10 is considered to be severe, while between 5 and 10 should warrant caution. None of the variable exceed a threshold of 10, indicating multicollinearity is not present.The highest VIF belongs to medv (median home value, VIF = 8.12), followed by rm (average rooms per dwelling, VIF = 5.81). These two variables are moderate and do not need requrie any removal. The remaining predictors fall below 5 and are acceptable levels of multicollinearity. There was strong pairwise correlation between "rad" and "tax" in the exploration section, howevr it is confirmed that multicollinearity does not exist in the dataset.

4. Linearity of Log-Odds (Box-Tidwell Test)

```
# Box-Tidwell test: add interaction of each continuous predictor with its log
# to test whether the log-odds relationship is linear
continuous_vars <- c(
    "zn", "indus", "nox", "rm", "age", "dis", "rad", "tax",
    "ptratio", "lstat", "medv"
)

# Create interaction terms x * log(x) for each continuous predictor
# (use log1p to handle zeros)
crime_bt <- crime_clean |>
    mutate(across(all_of(continuous_vars), ~ . * log1p(.), .names = "{.col}_logx"))

bt_formula <- as.formula(paste(
    "target ~",
    paste(continuous_vars, collapse = " + "),
    "+",
    paste(paste0(continuous_vars, "_logx"), collapse = " + ")
))

model_bt <- glm(bt_formula, data = crime_bt, family = binomial)

# Extract p-values for the interaction terms only
bt_results <- summary(model_bt)$coefficients |>
    as.data.frame() |>
    rownames_to_column("Term") |>
    filter(str_detect(Term, "_logx")) |>
    mutate(
        Variable = str_remove(Term, "_logx"),
        Significant = `Pr(>|z|)` < 0.05
```

```
    ) |>
    select(Variable, Estimate, `Pr(>|z|)`, Significant) |>
    mutate(across(where(is.numeric), ~ signif(., 3)))

bt_results |> knitr::kable(caption = "Box-Tidwell Test for Linearity of Log-Odds")
```

The Box-Tidwell test was used to assess whether each continuous predictor has a linear relationship with the log-odds of the outcome. A p-value greater than 0.05 violate the assumption. There are three variables that were flagged as significant:rm (p = 0.003), dis ($p < 0.001$), and tax ($p < 0.001$). This indicates that these predictors do not have strict lienear relationship with log-odds of high crime. Log transformation were applied to these variables in the data preparation section in order to be used in the model building. The remaining eight predictors did not have any evidence of violating the linearity assumption.

5 Build models

We will build models for two separate outcomes:

1. **target_flag** (binary): Whether the person was in a car crash (1) or not (0)
2. **target_amt** (continuous): The amount of money the crash cost if it occurs

For each outcome, we compare three models as per the textbook methods: - Model 1: Full model with all predictors - Model 2: Reduced model (removing highly correlated or insignificant variables) - Model 3: Backward stepwise selection model

5.1 Model 1. Full Logistic and Linear Models

As shown below, the most significant predictors for `target_flag` (whether a crash occurred) include variables related to the driver's profile and history. The AIC for this logistic model is `r\ AIC(model1_logistic)`.

Similarly, for `target_amt` (cost of crash), the full linear model identifies significant predictors. The R-squared for this model is `r\ summary(model1_linear)$r.squared`.

```
# Model 1: Full Logistic Model (target_flag)
# This model uses all available numeric predictors
model1_logistic <- glm(
    formula = target_flag ~ .,
    data = insurance_train_split |> select(-target_amt),
    family = binomial
)
```

```r
cat("\n=== Full Logistic Model Summary ===\n")
```

```
=== Full Logistic Model Summary ===
```

```r
summary(model1_logistic)
```

```
Call:
glm(formula = target_flag ~ ., family = binomial, data = select(insurance_train_split,
    -target_amt))

Coefficients: (1 not defined because of singularities)
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)               7.944e+00  4.389e+00   1.810 0.070287 .
kidsdriv                  4.407e-01  6.883e-02   6.403 1.52e-10 ***
age                      -7.728e-03  5.236e-03  -1.476 0.139948
homekids                  6.359e-03  4.222e-02   0.151 0.880264
yoj                       1.437e-02  1.258e-02   1.142 0.253363
income                   -1.855e-06  1.521e-06  -1.220 0.222471
home_val                  1.026e-06  1.940e-06   0.529 0.597085
travtime                  9.614e-03  6.062e-03   1.586 0.112719
bluebook                  5.090e-06  1.249e-05   0.407 0.683678
tif                      -5.336e-02  8.322e-03  -6.411 1.44e-10 ***
oldclaim                 -2.687e-05  1.382e-05  -1.945 0.051786 .
clm_freq                  1.617e-02  6.380e-02   0.253 0.799915
mvr_pts                   9.445e-02  1.587e-02   5.952 2.65e-09 ***
car_age                   4.851e-03  1.200e-02   0.404 0.685993
parent1_yes               4.088e-01  1.238e-01   3.301 0.000962 ***
mstatus_z_no              4.909e-01  9.584e-02   5.122 3.02e-07 ***
female                   -1.415e-01  1.278e-01  -1.108 0.268011
education_bachelors      -3.574e-01  1.326e-01  -2.694 0.007049 **
education_masters        -3.316e-01  2.080e-01  -1.594 0.110934
education_phd            -4.985e-01  2.513e-01  -1.984 0.047281 *
education_high_school     5.154e-02  1.091e-01   0.473 0.636525
job_doctor               -7.560e-01  3.343e-01  -2.261 0.023736 *
job_home_maker           -5.309e-01  1.844e-01  -2.879 0.003991 **
job_lawyer               -3.305e-01  2.103e-01  -1.572 0.116049
job_manager              -9.606e-01  1.614e-01  -5.951 2.67e-09 ***
job_professional         -2.989e-01  1.404e-01  -2.129 0.033285 *
```

```
job_student           -5.214e-01  1.722e-01  -3.028 0.002464 **
job_blue_collar       -1.258e-01  1.213e-01  -1.037 0.299751
job_na                -3.484e-01  2.242e-01  -1.554 0.120120
car_use_private       -7.139e-01  1.041e-01  -6.855 7.12e-12 ***
car_type_panel_truck   5.357e-01  1.873e-01   2.860 0.004237 **
car_type_pickup        5.797e-01  1.139e-01   5.091 3.57e-07 ***
car_type_sports_car    9.700e-01  1.497e-01   6.482 9.07e-11 ***
car_type_van           6.665e-01  1.439e-01   4.633 3.60e-06 ***
car_type_suv           7.805e-01  1.276e-01   6.118 9.46e-10 ***
red_car_yes           -1.006e-01  9.876e-02  -1.019 0.308307
revoked_yes            9.370e-01  1.048e-01   8.941  < 2e-16 ***
urbanicity_highly_rural -2.353e+00  1.278e-01 -18.416  < 2e-16 ***
age_missing            1.978e+00  1.230e+00   1.608 0.107754
car_age_missing        1.679e-01  1.368e-01   1.227 0.219674
home_val_missing       3.324e-01  9.554e-02   3.480 0.000502 ***
income_missing         1.810e-02  1.500e-01   0.121 0.903968
yoj_missing            6.266e-02  1.439e-01   0.435 0.663252
log_income            -7.355e-02  2.267e-02  -3.244 0.001177 **
log_home_val          -4.556e-01  3.831e-01  -1.189 0.234347
log_bluebook          -4.049e-01  1.392e-01  -2.908 0.003633 **
log_oldclaim           8.193e-02  1.980e-02   4.138 3.51e-05 ***
log_travtime           2.118e-01  1.727e-01   1.226 0.220103
claim_rate            -3.106e-06  3.730e-05  -0.083 0.933638
car_value_age_ratio    1.851e-05  1.850e-05   1.000 0.317098
risk_points_claims            NA         NA      NA       NA
age_bucket_older_driver  7.900e-01  1.886e-01   4.190 2.79e-05 ***
age_bucket_young_driver  4.971e-01  3.333e-01   1.492 0.135820
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 7531.0  on 6527  degrees of freedom
Residual deviance: 5748.1  on 6476  degrees of freedom
AIC: 5852.1

Number of Fisher Scoring iterations: 5
```

```
# Model 1: Full Linear Model (target_amt)
# This model uses all available numeric predictors
```

```r
model1_linear <- lm(
    formula = target_amt ~ .,
    data = insurance_train_split |> select(-target_flag)
)

cat("\n=== Full Linear Model Summary ===\n")
```

```
=== Full Linear Model Summary ===
```

```r
summary(model1_linear)
```

```

Call:
lm(formula = target_amt ~ ., data = select(insurance_train_split,
    -target_flag))

Residuals:
   Min     1Q Median     3Q    Max
 -5476  -1694   -760    387 103433

Coefficients: (1 not defined because of singularities)
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)              9.078e+02  7.454e+03   0.122  0.90307
kidsdriv                 3.108e+02  1.287e+02   2.414  0.01579 *
age                     -1.282e-01  9.262e+00  -0.014  0.98896
homekids                 3.280e+01  7.557e+01   0.434  0.66433
yoj                     -1.116e+00  2.189e+01  -0.051  0.95936
income                  -2.254e-03  2.672e-03  -0.843  0.39899
home_val                -6.342e-04  3.161e-03  -0.201  0.84098
travtime                 4.353e+00  1.018e+01   0.427  0.66908
bluebook                 9.426e-04  2.091e-02   0.045  0.96405
tif                     -4.269e+01  1.400e+01  -3.049  0.00230 **
oldclaim                -6.465e-02  2.699e-02  -2.396  0.01662 *
clm_freq                 1.338e+02  1.288e+02   1.038  0.29910
mvr_pts                  1.633e+02  3.062e+01   5.333 9.97e-08 ***
car_age                 -1.753e+01  2.066e+01  -0.848  0.39626
parent1_yes              5.603e+02  2.313e+02   2.422  0.01545 *
mstatus_z_no             5.819e+02  1.679e+02   3.467  0.00053 ***
female                  -3.282e+02  2.135e+02  -1.537  0.12437
```

```
education_bachelors     -2.888e+02  2.391e+02  -1.208  0.22723
education_masters        3.296e+00  3.498e+02   0.009  0.99248
education_phd           -2.056e+02  4.151e+02  -0.495  0.62046
education_high_school   -1.813e+02  1.997e+02  -0.908  0.36405
job_doctor              -9.151e+02  5.033e+02  -1.818  0.06908 .
job_home_maker          -4.712e+02  3.171e+02  -1.486  0.13727
job_lawyer              -4.518e+02  3.503e+02  -1.290  0.19716
job_manager             -1.158e+03  2.726e+02  -4.247 2.20e-05 ***
job_professional        -1.423e+02  2.485e+02  -0.573  0.56696
job_student             -4.002e+02  3.120e+02  -1.283  0.19962
job_blue_collar         -8.843e+01  2.221e+02  -0.398  0.69060
job_na                  -7.612e+02  3.922e+02  -1.941  0.05235 .
car_use_private         -8.037e+02  1.889e+02  -4.255 2.12e-05 ***
car_type_panel_truck     2.656e+02  3.251e+02   0.817  0.41401
car_type_pickup          2.608e+02  1.948e+02   1.339  0.18074
car_type_sports_car      1.009e+03  2.553e+02   3.952 7.85e-05 ***
car_type_van             5.800e+02  2.449e+02   2.369  0.01787 *
car_type_suv             6.435e+02  2.084e+02   3.088  0.00202 **
red_car_yes             -8.997e+01  1.724e+02  -0.522  0.60185
revoked_yes              5.471e+02  2.003e+02   2.731  0.00633 **
urbanicity_highly_rural -1.623e+03  1.617e+02 -10.038  < 2e-16 ***
age_missing              1.162e+03  2.095e+03   0.555  0.57892
car_age_missing          7.637e+01  2.433e+02   0.314  0.75362
home_val_missing         1.516e+02  1.741e+02   0.871  0.38383
income_missing           1.257e+02  2.608e+02   0.482  0.62983
yoj_missing              3.243e+02  2.523e+02   1.285  0.19869
log_income              -3.847e+01  4.027e+01  -0.955  0.33946
log_home_val            -1.993e+01  6.518e+02  -0.031  0.97561
log_bluebook             1.353e+02  2.453e+02   0.551  0.58139
log_oldclaim             2.420e+01  3.927e+01   0.616  0.53773
log_travtime             1.811e+02  2.828e+02   0.640  0.52189
claim_rate               1.335e-01  7.295e-02   1.830  0.06727 .
car_value_age_ratio      1.344e-02  3.263e-02   0.412  0.68031
risk_points_claims              NA         NA      NA       NA
age_bucket_older_driver  5.248e+02  3.444e+02   1.524  0.12758
age_bucket_young_driver  3.244e+02  6.563e+02   0.494  0.62111
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 4643 on 6476 degrees of freedom
Multiple R-squared:  0.0709,    Adjusted R-squared:  0.06359
F-statistic:  9.69 on 51 and 6476 DF,  p-value: < 2.2e-16
```

5.2 Model 2. Reduced Models (Addressing Multicollinearity)

In the Data Exploration section, we identified several variables with high correlations. For the reduced models, we remove highly correlated predictors while retaining theoretically important ones.

The AIC for the reduced logistic model is `r\ AIC(model2_logistic)`, and the R-squared for the reduced linear model is `r\ summary(model2_linear)$r.squared`.

```r
# Model 2: Reduced Logistic Model
# Keep only important predictors based on correlation analysis
reduced_vars <- c("kidsdriv", "age", "homekids", "yoj", "income", "home_val",
                  "travtime", "bluebook", "tif", "oldclaim", "clm_freq", "mvr_pts", "car_age")

model2_logistic <- glm(
    formula = as.formula(paste("target_flag ~", paste(reduced_vars, collapse = "+"))),
    data = insurance_train_split,
    family = binomial
)

cat("\n=== Reduced Logistic Model Summary ===\n")
```

```
=== Reduced Logistic Model Summary ===
```

```r
summary(model2_logistic)
```

```
Call:
glm(formula = as.formula(paste("target_flag ~", paste(reduced_vars,
    collapse = "+"))), family = binomial, data = insurance_train_split)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -4.658e-01  2.236e-01  -2.083 0.037259 *
kidsdriv     2.977e-01  6.039e-02   4.930 8.23e-07 ***
age         -8.546e-03  3.994e-03  -2.140 0.032383 *
homekids     8.324e-02  3.304e-02   2.519 0.011754 *
yoj         -1.752e-02  7.679e-03  -2.281 0.022524 *
```

```
income      -3.230e-06  1.133e-06  -2.851 0.004353 **
home_val    -7.670e-07  6.134e-07  -1.250 0.211143
travtime     8.276e-03  1.848e-03   4.478 7.52e-06 ***
bluebook    -1.400e-05  4.096e-06  -3.419 0.000629 ***
tif         -4.374e-02  7.562e-03  -5.784 7.28e-09 ***
oldclaim     6.127e-06  3.481e-06   1.760 0.078418 .
clm_freq     2.691e-01  2.868e-02   9.381  < 2e-16 ***
mvr_pts      1.479e-01  1.394e-02  10.616  < 2e-16 ***
car_age     -2.270e-02  6.078e-03  -3.735 0.000188 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 7531.0  on 6527  degrees of freedom
Residual deviance: 6803.1  on 6514  degrees of freedom
AIC: 6831.1

Number of Fisher Scoring iterations: 4
```

```r
# Model 2: Reduced Linear Model
model2_linear <- lm(
    formula = as.formula(paste("target_amt ~", paste(reduced_vars, collapse = "+"))),
    data = insurance_train_split
)

cat("\n=== Reduced Linear Model Summary ===\n")
```

```
=== Reduced Linear Model Summary ===
```

```r
summary(model2_linear)
```

```
Call:
lm(formula = as.formula(paste("target_amt ~", paste(reduced_vars,
    collapse = "+"))), data = insurance_train_split)

Residuals:
   Min      1Q Median      3Q     Max
 -4857  -1565   -968   -200 104001
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.517e+03  4.385e+02   3.459 0.000546 ***
kidsdriv     2.569e+02  1.300e+02   1.977 0.048115 *
age         -4.561e+00  7.936e+00  -0.575 0.565478
homekids     1.081e+02  6.768e+01   1.598 0.110143
yoj         -1.700e+01  1.568e+01  -1.084 0.278264
income      -2.187e-03  2.083e-03  -1.050 0.293755
home_val    -9.346e-04  1.117e-03  -0.836 0.402940
travtime     6.382e+00  3.664e+00   1.742 0.081575 .
bluebook     1.626e-02  7.721e-03   2.106 0.035246 *
tif         -3.865e+01  1.419e+01  -2.725 0.006450 **
oldclaim     5.048e-03  7.649e-03   0.660 0.509260
clm_freq     2.661e+02  6.168e+01   4.314 1.63e-05 ***
mvr_pts      2.290e+02  2.977e+01   7.692 1.66e-14 ***
car_age     -3.416e+01  1.169e+01  -2.923 0.003475 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4725 on 6514 degrees of freedom
Multiple R-squared:  0.03187,   Adjusted R-squared:  0.02994
F-statistic:  16.5 on 13 and 6514 DF,  p-value: < 2.2e-16
```

5.3 Model 3. Backward Stepwise Selection Models

For this model, we performed backward stepwise selection from the full model to identify the most important predictors.

The AIC for the stepwise logistic model is `r\ AIC(model3_logistic)`, and the R-squared for the stepwise linear model is `r\ summary(model3_linear)$r.squared`.

```r
# Model 3: Backward Stepwise Logistic Model
full_logistic <- glm(
    formula = target_flag ~ .,
    data = insurance_train_split |> select(-target_amt),
    family = binomial
)

model3_logistic <- step(full_logistic, direction = "backward", trace = 0)
```

```r
cat("\n=== Stepwise Logistic Model Summary ===\n")
```

```
=== Stepwise Logistic Model Summary ===
```

```r
summary(model3_logistic)
```

```
Call:
glm(formula = target_flag ~ kidsdriv + age + yoj + travtime +
    tif + oldclaim + mvr_pts + parent1_yes + mstatus_z_no + education_bachelors +
    education_masters + education_phd + job_doctor + job_home_maker +
    job_manager + job_professional + job_student + car_use_private +
    car_type_panel_truck + car_type_pickup + car_type_sports_car +
    car_type_van + car_type_suv + revoked_yes + urbanicity_highly_rural +
    age_missing + home_val_missing + log_income + log_home_val +
    log_bluebook + log_oldclaim + age_bucket_older_driver + age_bucket_young_driver,
    family = binomial, data = select(insurance_train_split, -target_amt))

Coefficients:
                         Estimate Std. Error z value Pr(>|z|)
(Intercept)             6.822e+00  1.506e+00   4.529 5.92e-06 ***
kidsdriv                4.458e-01  6.187e-02   7.205 5.82e-13 ***
age                    -8.248e-03  4.775e-03  -1.728 0.084068 .
yoj                     1.730e-02  1.193e-02   1.450 0.146972
travtime                1.651e-02  2.121e-03   7.785 6.95e-15 ***
tif                    -5.304e-02  8.281e-03  -6.405 1.50e-10 ***
oldclaim               -2.843e-05  5.244e-06  -5.422 5.89e-08 ***
mvr_pts                 9.366e-02  1.582e-02   5.922 3.18e-09 ***
parent1_yes             4.165e-01  1.129e-01   3.688 0.000226 ***
mstatus_z_no            4.828e-01  9.308e-02   5.187 2.14e-07 ***
education_bachelors    -4.629e-01  8.877e-02  -5.215 1.84e-07 ***
education_masters      -6.072e-01  1.005e-01  -6.043 1.51e-09 ***
education_phd          -8.093e-01  1.645e-01  -4.920 8.64e-07 ***
job_doctor             -4.626e-01  2.937e-01  -1.575 0.115221
job_home_maker         -4.048e-01  1.687e-01  -2.400 0.016413 *
job_manager            -7.864e-01  1.242e-01  -6.331 2.44e-10 ***
job_professional       -1.566e-01  1.105e-01  -1.418 0.156295
job_student            -4.131e-01  1.559e-01  -2.650 0.008059 **
car_use_private        -6.931e-01  8.427e-02  -8.225  < 2e-16 ***
```

```
car_type_panel_truck     6.116e-01  1.552e-01   3.941 8.11e-05 ***
car_type_pickup          5.774e-01  1.109e-01   5.207 1.91e-07 ***
car_type_sports_car      9.063e-01  1.226e-01   7.390 1.47e-13 ***
car_type_van             6.963e-01  1.362e-01   5.114 3.15e-07 ***
car_type_suv             7.223e-01  9.643e-02   7.491 6.85e-14 ***
revoked_yes              9.368e-01  1.044e-01   8.977  < 2e-16 ***
urbanicity_highly_rural -2.344e+00  1.273e-01 -18.412  < 2e-16 ***
age_missing              1.925e+00  1.220e+00   1.577 0.114814
home_val_missing         3.194e-01  8.744e-02   3.653 0.000259 ***
log_income              -8.588e-02  2.028e-02  -4.234 2.29e-05 ***
log_home_val            -3.525e-01  1.204e-01  -2.929 0.003402 **
log_bluebook            -3.376e-01  6.217e-02  -5.430 5.63e-08 ***
log_oldclaim             8.674e-02  1.106e-02   7.846 4.28e-15 ***
age_bucket_older_driver  8.352e-01  1.836e-01   4.550 5.36e-06 ***
age_bucket_young_driver  4.625e-01  3.260e-01   1.419 0.156018
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 7531.0  on 6527  degrees of freedom
Residual deviance: 5759.5  on 6494  degrees of freedom
AIC: 5827.5

Number of Fisher Scoring iterations: 5
```

```r
# Model 3: Backward Stepwise Linear Model
full_linear <- lm(
    formula = target_amt ~ .,
    data = insurance_train_split |> select(-target_flag)
)

model3_linear <- step(full_linear, direction = "backward", trace = 0)

cat("\n=== Stepwise Linear Model Summary ===\n")
```

```
=== Stepwise Linear Model Summary ===
```

```r
summary(model3_linear)
```

```

Call:
lm(formula = target_amt ~ kidsdriv + income + tif + oldclaim +
    clm_freq + mvr_pts + car_age + parent1_yes + mstatus_z_no +
    female + job_doctor + job_manager + car_use_private + car_type_sports_car +
    car_type_van + car_type_suv + revoked_yes + urbanicity_highly_rural +
    log_bluebook + log_travtime + claim_rate + age_bucket_older_driver,
    data = select(insurance_train_split, -target_flag))

Residuals:
   Min     1Q Median     3Q    Max
 -5661  -1702   -761    370 103447

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)             -2.607e+02  1.068e+03  -0.244 0.807097
kidsdriv                 3.507e+02  1.158e+02   3.030 0.002456 **
income                  -5.187e-03  1.519e-03  -3.415 0.000641 ***
tif                     -4.178e+01  1.393e+01  -3.000 0.002713 **
oldclaim                -7.157e-02  2.564e-02  -2.791 0.005273 **
clm_freq                 2.057e+02  6.904e+01   2.979 0.002900 **
mvr_pts                  1.723e+02  2.962e+01   5.817 6.28e-09 ***
car_age                 -3.336e+01  1.158e+01  -2.882 0.003970 **
parent1_yes              6.379e+02  2.014e+02   3.168 0.001543 **
mstatus_z_no             6.349e+02  1.370e+02   4.636 3.62e-06 ***
female                  -2.883e+02  1.702e+02  -1.694 0.090321 .
job_doctor              -6.149e+02  3.673e+02  -1.674 0.094111 .
job_manager             -9.248e+02  1.857e+02  -4.980 6.51e-07 ***
car_use_private         -8.311e+02  1.310e+02  -6.343 2.41e-10 ***
car_type_sports_car      9.203e+02  2.393e+02   3.846 0.000121 ***
car_type_van             4.405e+02  2.122e+02   2.076 0.037935 *
car_type_suv             5.438e+02  1.862e+02   2.920 0.003511 **
revoked_yes              5.487e+02  1.982e+02   2.769 0.005640 **
urbanicity_highly_rural -1.586e+03  1.569e+02 -10.106  < 2e-16 ***
log_bluebook             1.617e+02  1.073e+02   1.507 0.131773
log_travtime             2.974e+02  1.013e+02   2.935 0.003347 **
claim_rate               1.609e-01  6.197e-02   2.596 0.009440 **
```

```
age_bucket_older_driver  4.797e+02  3.047e+02   1.575 0.115396
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4638 on 6505 degrees of freedom
Multiple R-squared:  0.06857,   Adjusted R-squared:  0.06542
F-statistic: 21.77 on 22 and 6505 DF,  p-value: < 2.2e-16
```

Of these three models, the stepwise models have the lowest AIC for logistic and highest R-squared for linear. In the next section, we will evaluate the performance of these models to determine which ones are best for predicting the insurance outcomes.

6 Select Models

We evaluate both logistic and linear regression models using standard metrics from the textbook methods.

6.1 Evaluation Metrics Function

```
#' Evaluate Logistic Regression Models
#' @param model A glm logistic model
#' @param model_name Character name of the model
#' @param data Test dataframe
#' @param target_col Name of target column
#' @return Tibble with evaluation metrics
evaluate_logistic_model <- function(model, model_name, data, target_col = "target_flag") {
    probs <- predict(model, newdata = data, type = "response")
    preds <- factor(ifelse(probs > 0.5, 1, 0), levels = c(0, 1))
    actuals <- factor(data[[target_col]], levels = c(0, 1))

    cm <- confusionMatrix(preds, actuals, positive = "1")
    roc_obj <- roc(data[[target_col]], probs, quiet = TRUE)
    auc_val <- as.numeric(pROC::auc(roc_obj))

    tibble(
        Model = model_name,
        Deviance = model$deviance,
        Accuracy = cm$overall["Accuracy"],
        Error_Rate = 1 - cm$overall["Accuracy"],
        Precision = cm$byClass["Precision"],
```

```
        Sensitivity = cm$byClass["Sensitivity"],
        Specificity = cm$byClass["Specificity"],
        F1 = cm$byClass["F1"],
        AUC = auc_val,
        AIC = AIC(model),
        BIC = BIC(model)
    )
}

#' Evaluate Linear Regression Models
#' @param model An lm linear model
#' @param model_name Character name of the model
#' @param data Test dataframe
#' @param target_col Name of target column
#' @return Tibble with evaluation metrics
evaluate_linear_model <- function(model, model_name, data, target_col = "target_amt") {
    preds <- predict(model, newdata = data)
    actuals <- data[[target_col]]

    # Calculate metrics
    mse <- mean((actuals - preds)^2)
    rmse <- sqrt(mse)
    mae <- mean(abs(actuals - preds))
    r2 <- summary(model)$r.squared
    adj_r2 <- summary(model)$adj.r.squared

    tibble(
        Model = model_name,
        MSE = mse,
        RMSE = rmse,
        MAE = mae,
        R_Squared = r2,
        Adj_R_Squared = adj_r2,
        AIC = AIC(model),
        BIC = BIC(model)
    )
}

# Evaluate all logistic models
```

```r
cat("=== Evaluating Logistic Regression Models ===\n")
```

```
=== Evaluating Logistic Regression Models ===
```

```r
logistic_models <- list(
    "Full Logistic Model" = model1_logistic,
    "Reduced Logistic Model" = model2_logistic,
    "Stepwise Logistic Model" = model3_logistic
)

logistic_comparison <- imap_dfr(logistic_models, ~ {
    evaluate_logistic_model(.x, .y, insurance_test_split)
}) |> arrange(AIC)

# Display logistic model comparison
logistic_comparison |>
    knitr::kable(digits = 4, caption = "Logistic Model Comparison")
```

Table 2: Logistic Model Comparison

Model	Deviance	Accuracy	Error_Rate	Precision	Sensitivity	Specificity	F1	AUC	AIC	BIC
Stepwise Logistic Model	5759.498	0.7924	0.2076	0.6679	0.4282	0.9234	0.5219	0.8099	5827.498	6058.149
Full Logistic Model	5748.141	0.7961	0.2039	0.6800	0.4329	0.9267	0.5290	0.8104	5852.141	6204.901
Reduced Logistic Model	6803.144	0.7446	0.2554	0.5556	0.1736	0.9500	0.2646	0.6810	6831.144	6926.118

```r
# Select best logistic model
best_logistic <- logistic_comparison |> slice(1)
cat("\nBest Logistic Model:", best_logistic$Model, "\n")
```

```
Best Logistic Model: Stepwise Logistic Model
```

```r
# Evaluate all linear models
cat("\n=== Evaluating Linear Regression Models ===\n")
```

```
=== Evaluating Linear Regression Models ===
```

```r
linear_models <- list(
    "Full Linear Model" = model1_linear,
    "Reduced Linear Model" = model2_linear,
    "Stepwise Linear Model" = model3_linear
)

linear_comparison <- imap_dfr(linear_models, ~ {
    evaluate_linear_model(.x, .y, insurance_test_split)
}) |> arrange(desc(R_Squared))

# Display linear model comparison
linear_comparison |>
    knitr::kable(digits = 4, caption = "Linear Model Comparison")
```

Table 3: Linear Model Comparison

Model	MSE	RMSE	MAE	R_Squared	Adj_R_Squared	AIC	BIC
Full Linear Model	17184817	4145.457	1928.511	0.0709	0.0636	128811.8	129171.4
Stepwise Linear Model	17094366	4134.533	1926.207	0.0686	0.0654	128770.2	128933.0
Reduced Linear Model	17873538	4227.711	2028.093	0.0319	0.0299	129004.5	129106.2

```r
# Select best linear model
best_linear <- linear_comparison |> slice(1)
cat("\nBest Linear Model:", best_linear$Model, "\n")
```

```
Best Linear Model: Full Linear Model
```

For the logistic models, we use AIC as our primary criterion for model selection, with lower AIC indicating better model fit. The Stepwise Logistic Model achieved the lowest AIC of 5827.498296, with an accuracy of 0.792 and AUC of 0.81.

For the linear regression models, we use R-squared to select the best model, with higher R-squared indicating better model fit. The Full Linear Model achieved the highest R-squared of 0.071 with RMSE of 4145.46.

6.2 Selected Model Summaries

```r
# Display summary of best logistic model
cat("=== Best Logistic Model Summary ===\n")
```

```
=== Best Logistic Model Summary ===
```

```r
summary(logistic_models[[best_logistic$Model]])
```

```
Call:
glm(formula = target_flag ~ kidsdriv + age + yoj + travtime +
    tif + oldclaim + mvr_pts + parent1_yes + mstatus_z_no + education_bachelors +
    education_masters + education_phd + job_doctor + job_home_maker +
    job_manager + job_professional + job_student + car_use_private +
    car_type_panel_truck + car_type_pickup + car_type_sports_car +
    car_type_van + car_type_suv + revoked_yes + urbanicity_highly_rural +
    age_missing + home_val_missing + log_income + log_home_val +
    log_bluebook + log_oldclaim + age_bucket_older_driver + age_bucket_young_driver,
    family = binomial, data = select(insurance_train_split, -target_amt))

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)              6.822e+00  1.506e+00   4.529 5.92e-06 ***
kidsdriv                 4.458e-01  6.187e-02   7.205 5.82e-13 ***
age                     -8.248e-03  4.775e-03  -1.728 0.084068 .
yoj                      1.730e-02  1.193e-02   1.450 0.146972
travtime                 1.651e-02  2.121e-03   7.785 6.95e-15 ***
tif                     -5.304e-02  8.281e-03  -6.405 1.50e-10 ***
oldclaim                -2.843e-05  5.244e-06  -5.422 5.89e-08 ***
mvr_pts                  9.366e-02  1.582e-02   5.922 3.18e-09 ***
parent1_yes              4.165e-01  1.129e-01   3.688 0.000226 ***
mstatus_z_no             4.828e-01  9.308e-02   5.187 2.14e-07 ***
education_bachelors     -4.629e-01  8.877e-02  -5.215 1.84e-07 ***
education_masters       -6.072e-01  1.005e-01  -6.043 1.51e-09 ***
education_phd           -8.093e-01  1.645e-01  -4.920 8.64e-07 ***
job_doctor              -4.626e-01  2.937e-01  -1.575 0.115221
job_home_maker          -4.048e-01  1.687e-01  -2.400 0.016413 *
job_manager             -7.864e-01  1.242e-01  -6.331 2.44e-10 ***
job_professional        -1.566e-01  1.105e-01  -1.418 0.156295
job_student             -4.131e-01  1.559e-01  -2.650 0.008059 **
```

```
car_use_private         -6.931e-01  8.427e-02  -8.225  < 2e-16 ***
car_type_panel_truck     6.116e-01  1.552e-01   3.941 8.11e-05 ***
car_type_pickup          5.774e-01  1.109e-01   5.207 1.91e-07 ***
car_type_sports_car      9.063e-01  1.226e-01   7.390 1.47e-13 ***
car_type_van             6.963e-01  1.362e-01   5.114 3.15e-07 ***
car_type_suv             7.223e-01  9.643e-02   7.491 6.85e-14 ***
revoked_yes              9.368e-01  1.044e-01   8.977  < 2e-16 ***
urbanicity_highly_rural -2.344e+00  1.273e-01 -18.412  < 2e-16 ***
age_missing              1.925e+00  1.220e+00   1.577 0.114814
home_val_missing         3.194e-01  8.744e-02   3.653 0.000259 ***
log_income              -8.588e-02  2.028e-02  -4.234 2.29e-05 ***
log_home_val            -3.525e-01  1.204e-01  -2.929 0.003402 **
log_bluebook            -3.376e-01  6.217e-02  -5.430 5.63e-08 ***
log_oldclaim             8.674e-02  1.106e-02   7.846 4.28e-15 ***
age_bucket_older_driver  8.352e-01  1.836e-01   4.550 5.36e-06 ***
age_bucket_young_driver  4.625e-01  3.260e-01   1.419 0.156018
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 7531.0  on 6527  degrees of freedom
Residual deviance: 5759.5  on 6494  degrees of freedom
AIC: 5827.5

Number of Fisher Scoring iterations: 5
```

```r
# Display summary of best linear model
cat("\n=== Best Linear Model Summary ===\n")
```

```
=== Best Linear Model Summary ===
```

```r
summary(linear_models[[best_linear$Model]])
```

```
Call:
lm(formula = target_amt ~ ., data = select(insurance_train_split,
    -target_flag))

Residuals:
```

```
   Min     1Q Median     3Q    Max 
 -5476  -1694   -760    387 103433 

Coefficients: (1 not defined because of singularities)
                          Estimate Std. Error t value Pr(>|t|)    
(Intercept)              9.078e+02  7.454e+03   0.122  0.90307    
kidsdriv                 3.108e+02  1.287e+02   2.414  0.01579 *  
age                     -1.282e-01  9.262e+00  -0.014  0.98896    
homekids                 3.280e+01  7.557e+01   0.434  0.66433    
yoj                     -1.116e+00  2.189e+01  -0.051  0.95936    
income                  -2.254e-03  2.672e-03  -0.843  0.39899    
home_val                -6.342e-04  3.161e-03  -0.201  0.84098    
travtime                 4.353e+00  1.018e+01   0.427  0.66908    
bluebook                 9.426e-04  2.091e-02   0.045  0.96405    
tif                     -4.269e+01  1.400e+01  -3.049  0.00230 ** 
oldclaim                -6.465e-02  2.699e-02  -2.396  0.01662 *  
clm_freq                 1.338e+02  1.288e+02   1.038  0.29910    
mvr_pts                  1.633e+02  3.062e+01   5.333 9.97e-08 ***
car_age                 -1.753e+01  2.066e+01  -0.848  0.39626    
parent1_yes              5.603e+02  2.313e+02   2.422  0.01545 *  
mstatus_z_no             5.819e+02  1.679e+02   3.467  0.00053 ***
female                  -3.282e+02  2.135e+02  -1.537  0.12437    
education_bachelors     -2.888e+02  2.391e+02  -1.208  0.22723    
education_masters        3.296e+00  3.498e+02   0.009  0.99248    
education_phd           -2.056e+02  4.151e+02  -0.495  0.62046    
education_high_school   -1.813e+02  1.997e+02  -0.908  0.36405    
job_doctor              -9.151e+02  5.033e+02  -1.818  0.06908 .  
job_home_maker          -4.712e+02  3.171e+02  -1.486  0.13727    
job_lawyer              -4.518e+02  3.503e+02  -1.290  0.19716    
job_manager             -1.158e+03  2.726e+02  -4.247 2.20e-05 ***
job_professional        -1.423e+02  2.485e+02  -0.573  0.56696    
job_student             -4.002e+02  3.120e+02  -1.283  0.19962    
job_blue_collar         -8.843e+01  2.221e+02  -0.398  0.69060    
job_na                  -7.612e+02  3.922e+02  -1.941  0.05235 .  
car_use_private         -8.037e+02  1.889e+02  -4.255 2.12e-05 ***
car_type_panel_truck     2.656e+02  3.251e+02   0.817  0.41401    
car_type_pickup          2.608e+02  1.948e+02   1.339  0.18074    
car_type_sports_car      1.009e+03  2.553e+02   3.952 7.85e-05 ***
car_type_van             5.800e+02  2.449e+02   2.369  0.01787 *  
```

```
car_type_suv             6.435e+02  2.084e+02   3.088  0.00202 **
red_car_yes             -8.997e+01  1.724e+02  -0.522  0.60185
revoked_yes              5.471e+02  2.003e+02   2.731  0.00633 **
urbanicity_highly_rural -1.623e+03  1.617e+02 -10.038  < 2e-16 ***
age_missing              1.162e+03  2.095e+03   0.555  0.57892
car_age_missing          7.637e+01  2.433e+02   0.314  0.75362
home_val_missing         1.516e+02  1.741e+02   0.871  0.38383
income_missing           1.257e+02  2.608e+02   0.482  0.62983
yoj_missing              3.243e+02  2.523e+02   1.285  0.19869
log_income              -3.847e+01  4.027e+01  -0.955  0.33946
log_home_val            -1.993e+01  6.518e+02  -0.031  0.97561
log_bluebook             1.353e+02  2.453e+02   0.551  0.58139
log_oldclaim             2.420e+01  3.927e+01   0.616  0.53773
log_travtime             1.811e+02  2.828e+02   0.640  0.52189
claim_rate               1.335e-01  7.295e-02   1.830  0.06727 .
car_value_age_ratio      1.344e-02  3.263e-02   0.412  0.68031
risk_points_claims              NA         NA      NA       NA
age_bucket_older_driver  5.248e+02  3.444e+02   1.524  0.12758
age_bucket_young_driver  3.244e+02  6.563e+02   0.494  0.62111
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4643 on 6476 degrees of freedom
Multiple R-squared:  0.0709,    Adjusted R-squared:  0.06359
F-statistic:  9.69 on 51 and 6476 DF,  p-value: < 2.2e-16
```

To make inferences from our top performing models, we can look at the significance of the predictors.

For the logistic regression model (target_flag), the model identifies key factors that significantly predict whether a person will be in a car crash. The most important predictors include variables related to driving history, vehicle characteristics, and demographic factors.

For the linear regression model (target_amt), significant predictors help explain the variation in the cost of claims when crashes occur.

7 Results

Generate predictions on the evaluation dataset using the selected models:

```
# Get the best model names
best_logistic_name <- best_logistic$Model
best_linear_name <- best_linear$Model
```

```r
# Generate predictions for target_flag (probability of crash)
eval_probs <- predict(
    logistic_models[[best_logistic_name]],
    newdata = insurance_eval_prepared,
    type = "response"
)

# Generate predictions for target_amt (cost if crash)
eval_amounts <- predict(
    linear_models[[best_linear_name]],
    newdata = insurance_eval_prepared
)

# Create results dataframe
insurance_eval_results <- insurance_eval_prepared |>
    mutate(
        PREDICTED_target_flag = ifelse(eval_probs > 0.5, 1, 0),
        PREDICTED_target_amt = ifelse(PREDICTED_target_flag == 1,
                                      round(eval_amounts, 2),
                                      0)
    )

# Write predictions to CSV
write_csv(insurance_eval_results, "insurance_evaluation_predictions.csv")

# Display sample predictions
cat("Sample predictions for evaluation data:\n")
```

```
Sample predictions for evaluation data:
```

```r
insurance_eval_results |>
    select(PREDICTED_target_flag, PREDICTED_target_amt) |>
    head(10) |>
    knitr::kable(caption = "Sample Insurance Predictions")
```

Table 4: Sample Insurance Predictions

PREDICTED_target_flag	PREDICTED_target_amt
1	1392.17
1	2220.30

PREDICTED_target_flag	PREDICTED_target_amt
1	1347.85
1	2238.33
1	1129.25
0	0.00
1	2845.22
0	0.00
0	0.00
1	2029.34

We generate predictions for both outcomes:

1. **target_flag**: Whether the person will be in a car crash (binary: 0 or 1)
2. **target_amt**: The cost if a crash occurs (continuous dollar amount)

The predictions are written to `insurance_evaluation_predictions.csv` for submission.

8 Conclusion

In conclusion, our `best_model_info$Model` model, which was selected based on deviance and other performance metrics, demonstrates strong predictive capabilities for classifying neighborhoods as high-crime or low-crime based on a variety of environmental and structural factors. The significant predictors identified in the model provide valuable insights into the underlying drivers of crime in urban areas, highlighting the importance of air pollution, school crowding, aging infrastructure, and proximity to employment centers. These findings can inform policymakers and community leaders in their efforts to target interventions and allocate resources effectively to reduce crime rates. Future work could explore additional modeling techniques, such as regularization or ensemble methods, to further enhance predictive performance and interpretability.